

DYNAMICAL SYSTEMS MIDTERM PREPARATION

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EXAM PRACTICE QUESTIONS (CH 1–6.4)

Instructions. Read each question carefully to make sure you’ve answered it fully. You must show your work and explain your reasoning to receive full credit. No calculators or devices of any kind are allowed.

1) Consider the one-dimensional flow defined by

$$u' = r + u - u^3.$$

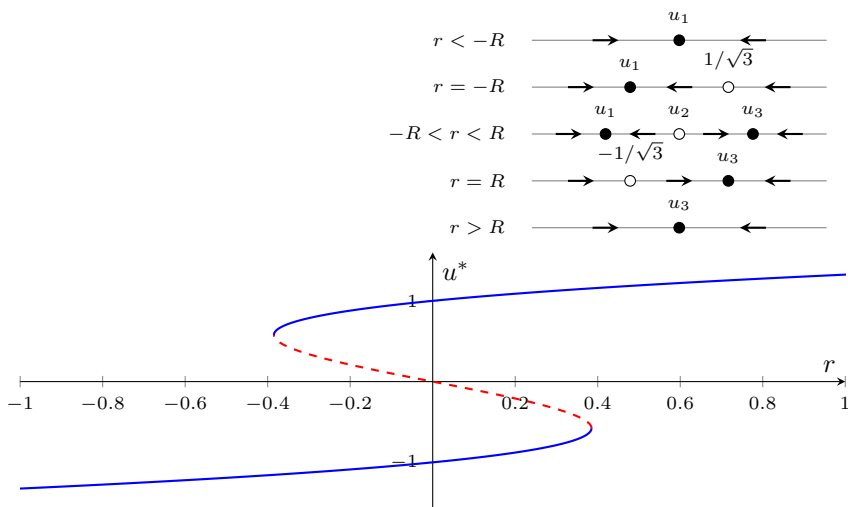
- (a) Sketch all the qualitatively different vector fields that occur as the parameter r is varied. Find the value(s) of r at which bifurcations occur, and classify them as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
- (b) Sketch a bifurcation diagram of fixed points u^* as a function of the parameter r . Be sure to indicate stability on your diagram.

Solution. Write $f(u; r) = r + u - u^3$. Equilibria satisfy $r = u^3 - u$, and

$$f_u = 1 - 3u^2, \quad f = f_u = 0 \iff (u, r) = \left(\pm \frac{1}{\sqrt{3}}, \mp \frac{2}{3\sqrt{3}} \right).$$

Let $R = 2/(3\sqrt{3})$. Since $f_r = 1$ and $f_{uu} = -6u \neq 0$ at either collision, both are **saddle-node bifurcations**. For $|r| < R$ the three ordered roots $u_1 < u_2 < u_3$ have stability attracting, repelling, attracting; for $|r| > R$ there is a single attracting root. At $r = -R$ the double positive root attracts from the right only; at $r = R$ the double negative root attracts from the left only. The other root is attracting.

In the diagrams, solid blue branches are attracting and dashed red branches are repelling. Filled phase-line dots are attracting; open dots are repelling or nonhyperbolic (as specified in the text). Arrows give the direction of increasing time.



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2) Consider the one-dimensional flow defined by

$$u' = r(u - 1) - u^2(u - 1).$$

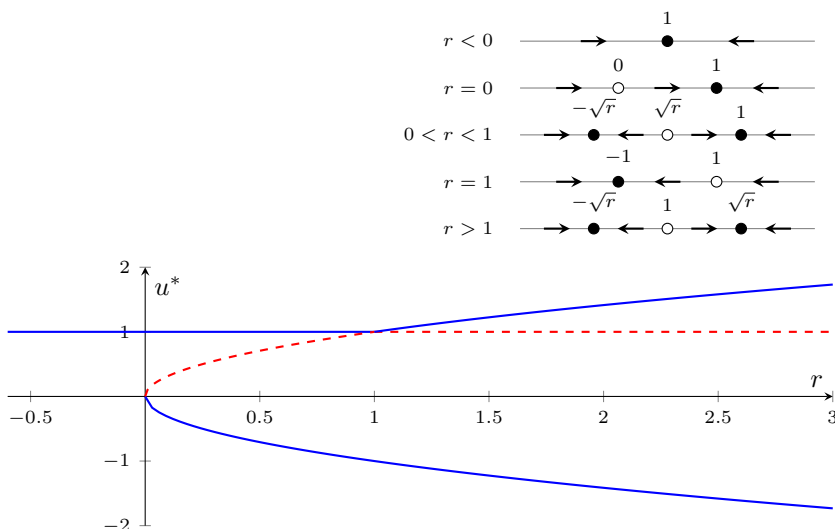
Date: October 1, 2026.

- (a) Sketch all the qualitatively different vector fields that occur as the parameter r is varied. Find the value(s) of r at which bifurcations occur, and classify them as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
- (b) Sketch a bifurcation diagram of fixed points u^* as a function of the parameter r . Be sure to indicate stability on your diagram.

Solution. Factor $f(u; r) = (u - 1)(r - u^2)$. The branches and their derivatives are

u^*	$f_u(u^*; r)$	stability
1	$r - 1$	attracting for $r < 1$, repelling for $r > 1$
$-\sqrt{r}$	$-2\sqrt{r}(\sqrt{r} + 1)$	attracting for $r > 0$
$\sqrt{r}$	$2\sqrt{r}(1 - \sqrt{r})$	repelling for $0 < r < 1$, attracting for $r > 1$

At $(u, r) = (0, 0)$, $f_r = -1$ and $f_{uu} = 2$, so a **saddle-node** creates the two square-root branches. At $(1, 1)$, putting $u = 1 + v$, $r = 1 + \mu$ gives $v' = v(\mu - 2v - v^2)$: a **transcritical** exchange of stability. At $r = 0$, $u = 0$ attracts from the left only; at $r = 1$, $u = 1$ attracts from the right only.



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- 3) The following equation describe motion on a line.

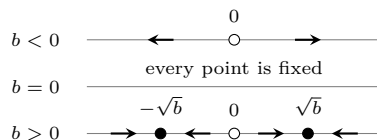
$$x' = bx(b - x^2).$$

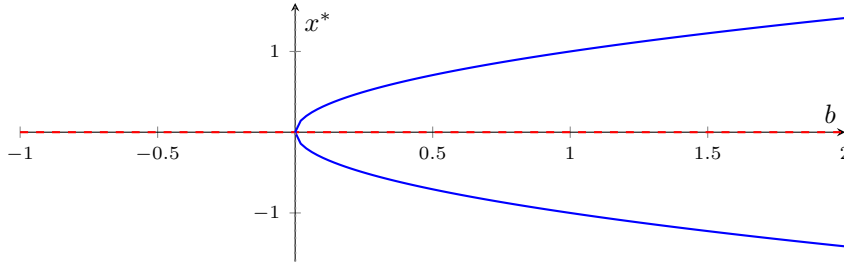
- (a) Sketch all the qualitatively different vector fields that occur as b is varied. Find any values of b at which bifurcations occur, and classify these as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
- (b) Sketch the bifurcation diagram of fixed points x^* vs. b , labeling the stability of each fixed point.
- (c) Describe the basin of attraction(s) for the stable fixed point(s) that exist when $b = 1$.

Solution. Here $f(x; b) = b^2x - bx^3$. For $b \neq 0$, the origin has $f_x(0; b) = b^2 > 0$, hence is **repelling on both sides of the parameter value zero**. For $b > 0$, the additional equilibria $\pm\sqrt{b}$ satisfy $f_x = -2b^2 < 0$ and are attracting. For $b < 0$ there are no other equilibria.

At $b = 0$, $f \equiv 0$: *every real number is an equilibrium*, each Lyapunov stable but none asymptotically stable. Thus $b = 0$ is a degenerate bifurcation, **not one of the four generic normal forms listed in the question**. The square-root branches resemble a supercritical pitchfork, but a generic pitchfork would change the stability of the central branch and would not have an entire line of equilibria at its critical parameter.

For $b = 1$, the basins of -1 and 1 are respectively $(-\infty, 0)$ and $(0, \infty)$; $x = 0$ stays fixed.





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- 4) The following equation describe motion on a line.

$$\frac{dx}{dt} = (r - 1 + x)(e^x - r).$$

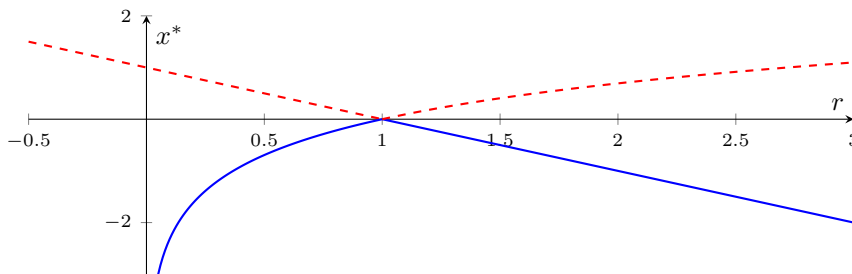
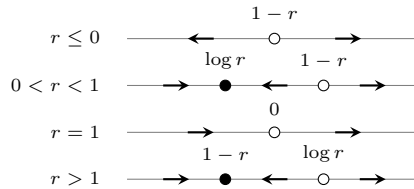
- Sketch all the qualitatively different vector fields that occur as b is varied. Find any values of b at which bifurcations occur, and classify these as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
- Sketch the bifurcation diagram of fixed points x^* vs. b , labeling the stability of each fixed point.
- Describe the basin of attraction(s) for the stable fixed point(s) that exist when $b = 1$.

Solution. The subparts use b , but the equation uses r ; interpret b as r throughout. The equilibrium branches are $x_1 = 1 - r$ for every r , and $x_2 = \log r$ for $r > 0$. At these points,

$$f_x(x_1; r) = e^{1-r} - r, \quad f_x(x_2; r) = r(r - 1 + \log r).$$

Both expressions vanish only at $r = 1$. For $0 < r < 1$, $x_2 < x_1$, with x_2 attracting and x_1 repelling. For $r > 1$, $x_1 < x_2$, and the stability reverses. With $\mu = r - 1$, the leading terms near zero are $x' = x^2 - \mu^2 +$ higher-order terms; shifting to the persistent branch $z = x + \mu$ gives $z' = z(z - 2\mu) +$ higher-order terms. This is a **transcritical bifurcation at $r = 1$** .

For $r \leq 0$, only the repelling equilibrium $1 - r$ remains. The logarithmic branch escapes to $-\infty$ as $r \downarrow 0$; this is not a finite local bifurcation. At $r = 1$, $x' = x(e^x - 1) > 0$ for $x \neq 0$. Therefore zero attracts from the left and repels from the right: it is semistable, not a two-sided stable fixed point. Its set of forward-convergent initial values is $(-\infty, 0]$; there is no asymptotically stable fixed point at this parameter.



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- 5) Consider the system $\dot{x} = ax - x^2 + x^3$.

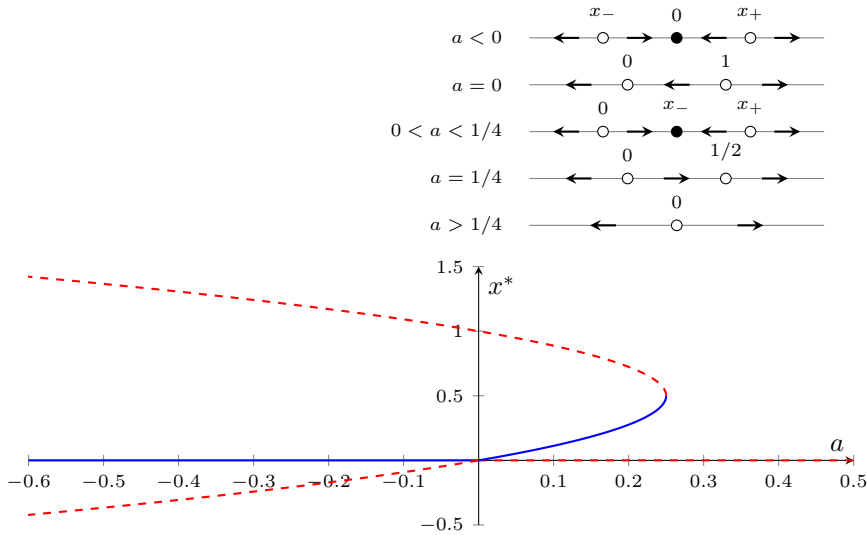
- Sketch all the qualitatively different vector fields that occur as a is varied. Find the values of a at which bifurcations occur, and classify these as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
- Sketch the bifurcation diagram of fixed points x^* vs. a .
- Can hysteresis occur in this system? Explain why or why not.

Solution. Factor $f = x(x^2 - x + a)$. Besides zero, equilibria are

$$x_{\pm} = \frac{1 \pm \sqrt{1 - 4a}}{2} \quad (a \leq 1/4), \quad f_x(0) = a, \quad f_x(x_{\pm}) = x_{\pm}(2x_{\pm} - 1).$$

For $a < 0$, $x_- < 0 < x_+$, the outer roots repel and zero attracts. For $0 < a < 1/4$, $0 < x_- < x_+$, zero and x_+ repel while x_- attracts. At $a = 0$, $f = x^2(x - 1)$, so zero attracts from the right only. The branches zero and x_- exchange stability in a **transcritical bifurcation**. At $a = 1/4$, $f = x(x - 1/2)^2$ and $x = 1/2$ attracts from the left only. Since $f_a = 1/2$ and $f_{xx} = 1$ there, this is a **saddle-node**. For $a > 1/4$, only the repelling origin remains.

There is **no standard quasistatic hysteresis loop between attracting equilibria**: no parameter interval has two attracting equilibria. Loss of the attracting branch at $1/4$ instead permits escape toward $+\infty$; it does not produce a second stable state and a return switch.



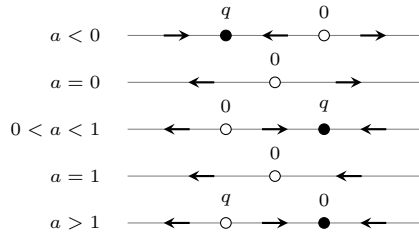
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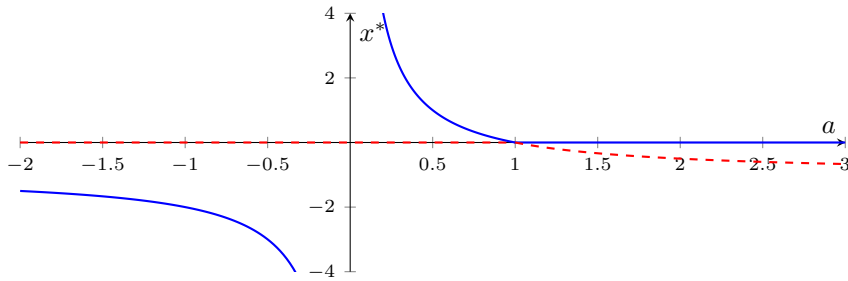
6) Consider the system $\dot{x} = x - a(x^2 + x)$.

(a) Sketch all the qualitatively different vector fields that occur as a is varied. Find the value(s) of a at which bifurcations occur, and classify them as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.

(b) Sketch the bifurcation diagram of fixed points x^* vs. a .

Solution. We have $f = x[(1 - a) - ax]$. Zero is always fixed; for $a \neq 0$ the other branch is $q = 1/a - 1$. Their derivatives are $f_x(0) = 1 - a$ and $f_x(q) = a - 1$. Thus zero repels for $a < 1$ and attracts for $a > 1$, while q has the opposite stability. At $a = 1$, $f = -x^2$, so zero attracts from the right only. Setting $\mu = a - 1$ gives $x' = -\mu x - (1 + \mu)x^2$, a **transcritical bifurcation at $a = 1$** . At $a = 0$, $x' = x$; the nonzero branch has escaped to infinity, not collided at a finite point.





□

7) For the following system of differential equations,

$$\mathbf{x}' = \begin{bmatrix} 2 & \alpha^2 \\ -1 & -4 \end{bmatrix} \mathbf{x}$$

where $\alpha > 0$ is an unspecified constant,

- determine the eigenvalues of the coefficient matrix in terms of α ,
- find the critical values of α at which the qualitative nature of the phase portrait experiences a dramatic change, and
- sketch the phase portrait for the different choices of α : one just below the each critical value and one just above each critical value.

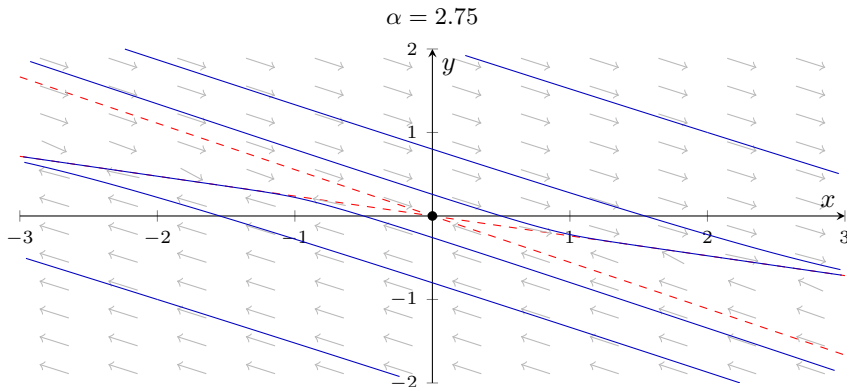
Solution. The trace is -2 , the determinant is $\alpha^2 - 8$, and

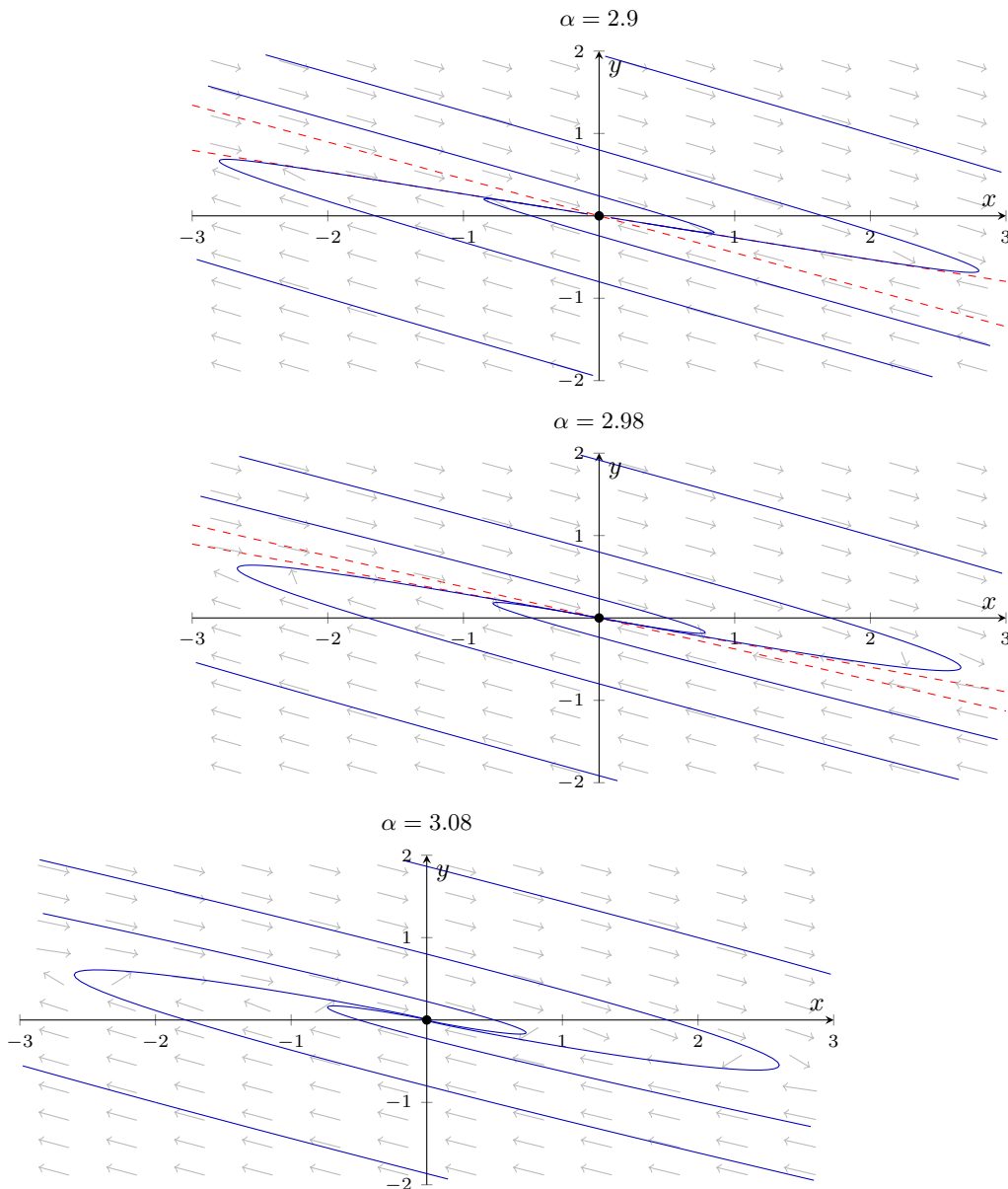
$$\lambda_{\pm} = -1 \pm \sqrt{9 - \alpha^2}, \quad v_{\pm} = \begin{pmatrix} 4 + \lambda_{\pm} \\ -1 \end{pmatrix}$$

are eigenvalues and corresponding eigenvectors when the eigenvalues are distinct. Thus

$0 < \alpha < \sqrt{8}$	saddle: $\lambda_- < 0 < \lambda_+$
$\alpha = \sqrt{8}$	line of equilibria $x = -4y$; transverse decay at rate 2
$\sqrt{8} < \alpha < 3$	attracting node, two distinct real eigenvalues
$\alpha = 3$	defective attracting node, repeated eigenvalue -1
$\alpha > 3$	attracting clockwise focus, $-1 \pm i\sqrt{\alpha^2 - 9}$

At $\alpha = 3$ the eigenspace is spanned by $(3, -1)^T$, and $(1, 0)^T$ is a generalized eigenvector since $(A+I)(1, 0)^T = (3, -1)^T$. The requested critical values are $\sqrt{8}$ and 3. Only the former loses hyperbolicity; the latter is a node-to-focus change. Clockwise rotation follows from $y' = -x < 0$ on the positive x -axis. The four plots sample just below and above each critical value. Gray arrows show the vector field; blue curves are representative trajectories; dashed lines are eigendirections when real.





□

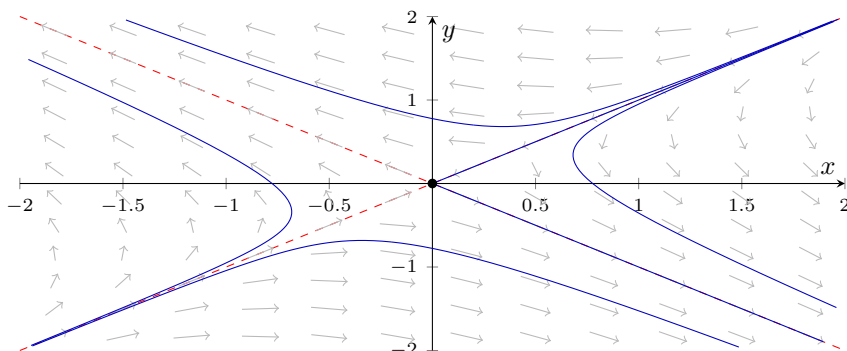
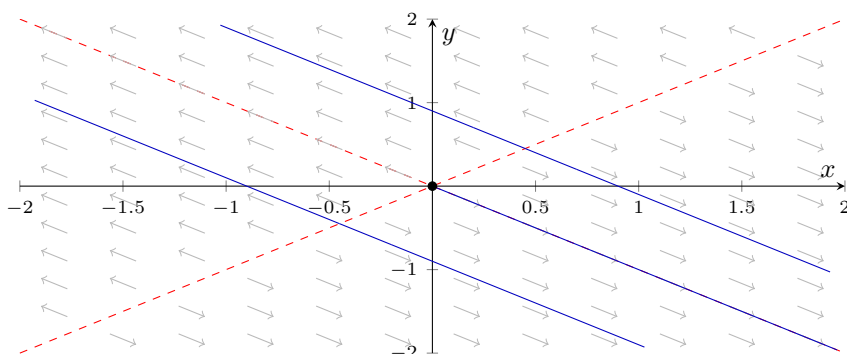
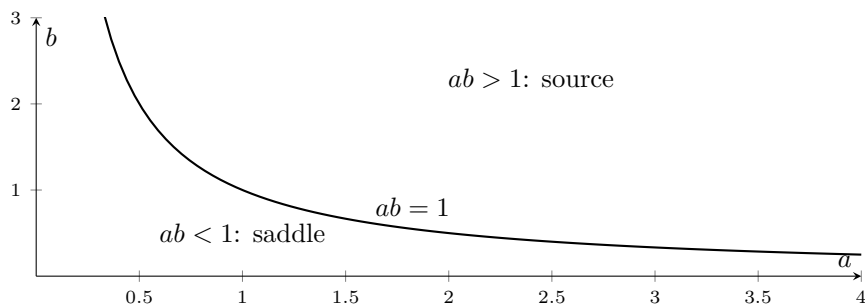
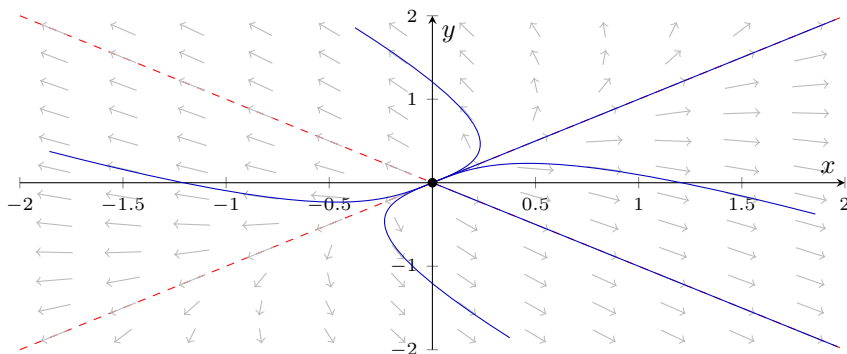
- 8) Consider the system 2D linear system $\dot{x} = Ax$, where $A = \begin{bmatrix} a & -1 \\ -1 & b \end{bmatrix}$ for parameters $a > 0$ and $b > 0$.
- Sketch a phase portrait and classify the fixed point at the origin for all possible qualitatively distinct cases as a function of parameters. Be sure to include eigenvectors associated with stable/unstable directions in your sketch as appropriate.
 - Summarize these results by indicating the regions that you've identified in the first quadrant of the (a, b) -plane.

Solution. Interpret the printed $\dot{x} = Ax$ as the differential equation $\dot{\mathbf{x}} = A\mathbf{x}$. Since A is symmetric,

$$T = a + b > 0, \quad D = ab - 1, \quad \lambda_{\pm} = \frac{a + b \pm \sqrt{(a - b)^2 + 4}}{2}, \quad v_{\pm} = \begin{pmatrix} 1 \\ a - \lambda_{\pm} \end{pmatrix}.$$

The eigenvalues are always distinct and real and $v_+ \perp v_-$. For $ab < 1$, $\lambda_- < 0 < \lambda_+$, giving a **saddle** with stable line $\mathbb{R}v_-$ and unstable line $\mathbb{R}v_+$. For $ab > 1$, both eigenvalues are positive, giving a **repelling node**. Its two invariant eigendirections both point away from zero.

On $ab = 1$, the eigenvalues are 0 and $a + b$. The entire line $y = ax$ consists of equilibria; transverse to it solutions grow exponentially. Thus the origin is nonisolated and unstable. In orthonormal eigenvector coordinates (ξ, η) , the exact general solution is $\xi(t) = \xi_0 e^{\lambda_- t}$, $\eta(t) = \eta_0 e^{\lambda_+ t}$, with ξ constant on the boundary. The following representative portraits use $a = b$; red dashed lines are the eigenvectors $v_- = (1, 1)^T$, $v_+ = (1, -1)^T$.

Saddle: $a = b = 1/2$ Line of equilibria: $a = b = 1$ Repelling node: $a = b = 2$ 

□

9) Consider the following planar dynamical system:

$$\begin{aligned}\dot{x} &= y - y^3 \\ \dot{y} &= -x - \frac{1}{2}y^2\end{aligned}$$

- (a) Find all fixed points and classify them using eigenvalues of the linearized system. For fixed points with real eigenvalues, find the eigenvectors as well.
 (b) Sketch the nullclines and a plausible phase portrait for this system.

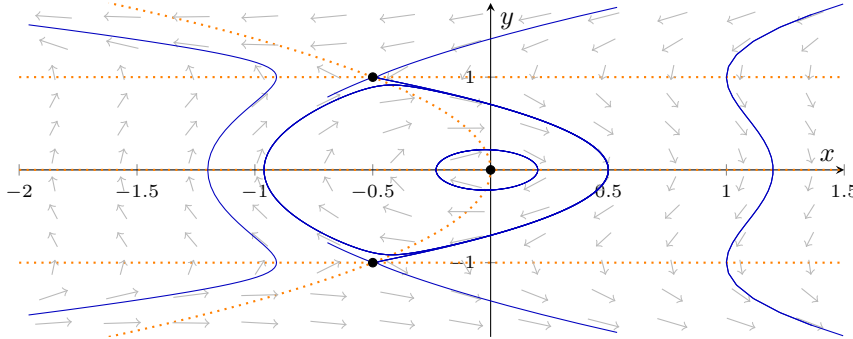
Solution. The nullclines are $y = 0, \pm 1$ and $x = -y^2/2$. Their intersections are $(0, 0)$ and $(-1/2, \pm 1)$. The Jacobian is

$$J(x, y) = \begin{pmatrix} 0 & 1 - 3y^2 \\ -1 & -y \end{pmatrix}.$$

At $(0, 0)$ the eigenvalues are $\pm i$: the linearization is a center, but these eigenvalues alone do not classify the nonlinear equilibrium. Here the nonlinear system is reversible under $(x, y, t) \mapsto (x, -y, -t)$. Near the origin its angular velocity is $-1 + O(\sqrt{x^2 + y^2})$, so each nearby orbit crosses the symmetry line $y = 0$ transversely. Reflection and time reversal at successive crossings, together with uniqueness, close the orbit. Thus the origin is a **nonlinear center**, surrounded by small clockwise periodic orbits.

At $(-1/2, 1)$ the eigenpairs are $\lambda = 1$, $v = (-2, 1)^T$ (unstable), and $\lambda = -2$, $v = (1, 1)^T$ (stable). At $(-1/2, -1)$ they are $\lambda = 2$, $v = (-1, 1)^T$ (unstable), and $\lambda = -1$, $v = (2, 1)^T$ (stable). Both points are **saddles**.

For the global sketch, $x' > 0$ for $y < -1$ or $0 < y < 1$, and $x' < 0$ for $-1 < y < 0$ or $y > 1$. Also $y' > 0$ to the left of $x = -y^2/2$ and $y' < 0$ to its right. Orange dotted curves are nullclines; numerical trajectories and the direction field give a plausible portrait.



10) Consider the following planar dynamical system:

$$\begin{aligned}\dot{x} &= x - xy \\ \dot{y} &= -y + xy - y^2\end{aligned}$$

- (a) Find all fixed points and classify them using eigenvalues of the linearized system. For fixed points with real eigenvalues, find the eigenvectors as well.
 (b) Sketch the nullclines and a plausible phase portrait for this system.

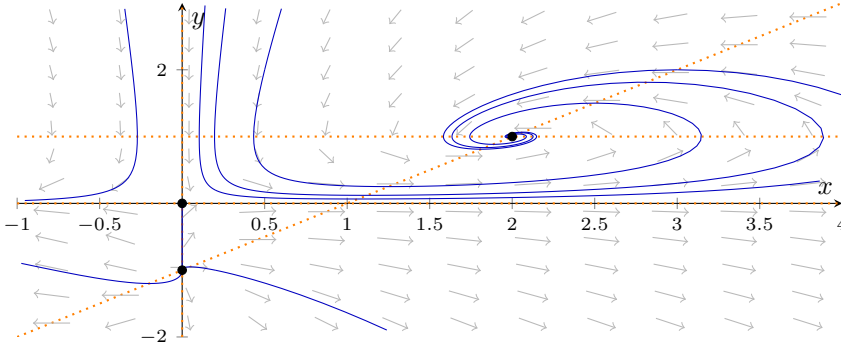
Solution. The nullclines are $x = 0$ or $y = 1$, and $y = 0$ or $y = x - 1$. Hence the fixed points are $(0, 0)$, $(0, -1)$, and $(2, 1)$. Since

$$J = \begin{pmatrix} 1 - y & -x \\ y & x - 1 - 2y \end{pmatrix},$$

we obtain:

- At $(0, 0)$: $\lambda = 1$ with $v = (1, 0)^T$, and $\lambda = -1$ with $v = (0, 1)^T$; a saddle.
- At $(0, -1)$: $\lambda = 2$ with $v = (1, -1)^T$, and $\lambda = 1$ with $v = (0, 1)^T$; a repelling node.
- At $(2, 1)$: $\lambda = (-1 \pm i\sqrt{7})/2$; an attracting focus. Rotation is counterclockwise because just to its right, $x' = 0$ and $y' > 0$.

The signs $\text{sgn } x' = \text{sgn}[x(1 - y)]$ and $\text{sgn } y' = \text{sgn}[y(x - 1 - y)]$ determine the arrows in every nullcline region. Both coordinate axes are invariant. The orange dotted nullclines, gray direction field, and blue trajectories give the portrait below.



□

11) Consider the system

$$\dot{x} = x^2 - y, \quad \dot{y} = x - y.$$

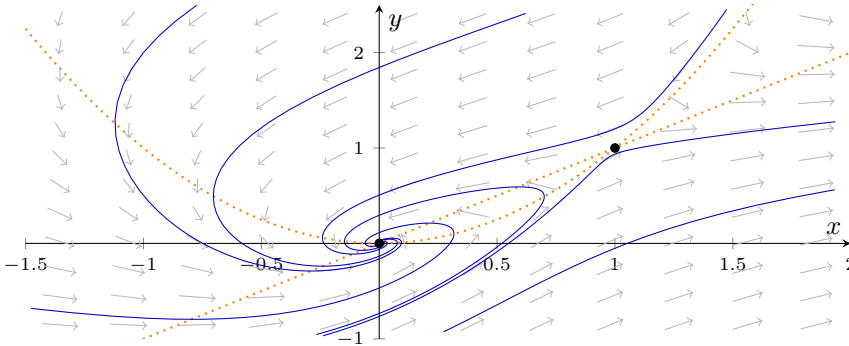
- (a) Find and classify all fixed points of this system, including eigenvalues and eigenvectors.
 (b) Sketch the nullclines and a plausible phase portrait for this system.

Solution. At equilibrium $y = x^2 = x$, so the fixed points are $(0,0)$ and $(1,1)$. The Jacobian is

$$J = \begin{pmatrix} 2x & -1 \\ 1 & -1 \end{pmatrix}.$$

At the origin, $\lambda_{\pm} = (-1 \pm i\sqrt{3})/2$, with complex eigenvectors $(1, -\lambda_{\pm})^T$. It is an attracting counterclockwise focus. At $(1,1)$, $\lambda_{\pm} = (1 \pm \sqrt{5})/2$, with eigenvectors $(1, 2 - \lambda_{\pm})^T$. Since $\lambda_- < 0 < \lambda_+$, it is a saddle; the corresponding eigenvectors are tangent to its stable and unstable manifolds.

The nullclines are the parabola $y = x^2$ and the line $y = x$. Below the parabola $x' > 0$, above it $x' < 0$; below the line $y' > 0$, above it $y' < 0$. These signs, the inward spiral at zero, and the saddle directions determine the sketch.



□

12) Consider the system

$$\dot{x} = x - x^3, \quad \dot{y} = -y.$$

- (a) Find and classify all fixed points of this system, including eigenvalues and eigenvectors.
 (b) Sketch the nullclines and a plausible phase portrait for this system.

Solution. The fixed points are $(0,0)$ and $(\pm 1,0)$, with

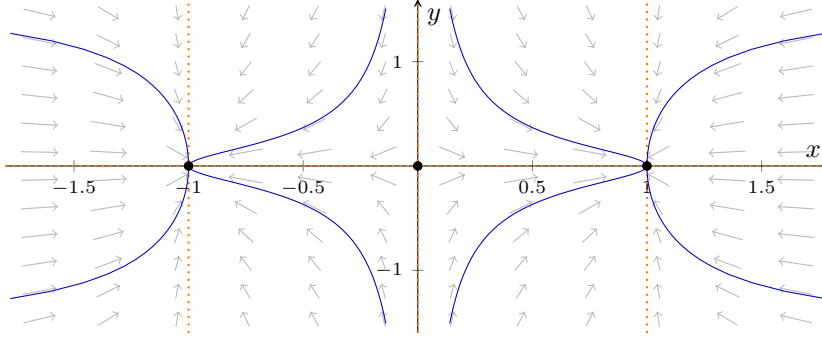
$$J = \text{diag}(1 - 3x^2, -1).$$

At zero the eigenpairs are $(1, (1,0)^T)$ and $(-1, (0,1)^T)$, giving a saddle. At both outer points the eigenpairs are $(-2, (1,0)^T)$ and $(-1, (0,1)^T)$, giving attracting nodes. The nullclines are $x = -1, 0, 1$ and $y = 0$.

Since $y(t) = y_0 e^{-t}$, all vertical components decay. For $x_0 \neq 0$ the scalar solution is

$$x(t) = \frac{\text{sgn}(x_0)}{\sqrt{1 + (x_0^{-2} - 1)e^{-2t}}},$$

so the right half-plane tends to $(1,0)$ and the left half-plane to $(-1,0)$. The entire y -axis is the stable manifold of the saddle; its unstable manifold lies on the x -axis between the two nodes.



□

- 13) An ODE system models the evolution of social distancing norms during a pandemic. The model assumes individuals reduce social contacts when infection rates are high and increase social contacts when infection rates are low. The model has two variables: the level of norm adherence in the community (x) and the proportion of infected persons in the community (y). The non-dimensionalized equations are shown below:

$$x' = ay + b - dx$$

$$y' = r(1-x)y(1-y) - y$$

where $x \in (0, \infty)$ and $y \in [0, 1)$. Assume $a \geq 1$, $b > 0$, $d > 0$ and $r > 0$.

- (a) Show the model has a unique disease-free equilibrium (DFE) $(x^*, 0)$ with $x^* > 0$.

- (b) Show there is always at least one endemic equilibrium (EE) $(x^*, y^*) \in \mathbb{R}^2$ with $x^*, y^* > 0$

Hint: solve for the non-zero equilibrium in x . This should yield a quadratic equation of the form $\alpha x^2 + \beta x + \gamma = 0$. Analyze the coefficients then state conditions for positive roots.

- (c) Suppose $b < d$. Show that the DFE is locally stable if $r < r_c$. What is r_c ?

Solution. (a) Setting $y = 0$ gives $x' = b - dx$, so the unique disease-free equilibrium is $(b/d, 0)$, with $b/d > 0$.

- (b) For an equilibrium with $y \neq 0$,

$$y = \frac{dx - b}{a}, \quad (1-x)(1-y) = \frac{1}{r}.$$

Eliminating y gives

$$Q(x) = dx^2 - (a + b + d)x + a + b - \frac{a}{r} = 0.$$

Its discriminant and roots are

$$\Delta = (a + b - d)^2 + \frac{4ad}{r} > 0, \quad x_{\pm} = \frac{a + b + d \pm \sqrt{\Delta}}{2d}, \quad y_{\pm} = \frac{dx_{\pm} - b}{a}.$$

The sum of the roots is positive. Both roots are positive when $r > a/(a + b)$; at equality the smaller root is zero; below it the smaller root is negative. Positivity of x alone does not imply positivity of y .

In fact, $Q(1) = Q((a + b)/d) = -a/r < 0$. Since Q opens upward,

$$x_+ > \max\{1, (a + b)/d\}, \quad y_+ > 1.$$

Consequently there is always a positive equilibrium in $\mathbb{R}^2$, as the literal statement of (b) requests. **However, it is outside the stated physical domain** $0 \leq y < 1$. Calling it an endemic equilibrium of that constrained model is misleading.

For a physical endemic equilibrium, $0 < y < 1$ implies $x < 1$, and $x = (ay + b)/d > b/d$. Thus necessarily $b < d$. On $b/d < x < \min\{1, (a + b)/d\}$ the function

$$F(x) = \frac{(1-x)(a + b - dx)}{a}$$

is strictly decreasing from $1 - b/d$ to zero. Hence there is exactly one physical endemic equilibrium if and only if

$$b < d \quad \text{and} \quad r > r_c := \frac{d}{d - b}.$$

It is the smaller-root equilibrium (x_-, y_-) . At $r = r_c$ it merges with the DFE, so it is not endemic there.

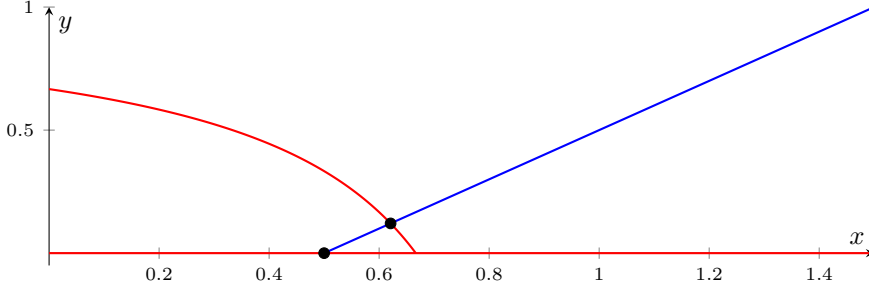
(c) At the DFE,

$$J = \begin{pmatrix} -d & a \\ 0 & r(1 - b/d) - 1 \end{pmatrix}.$$

For $b < d$, its eigenvalues are $-d$ and $r(1 - b/d) - 1$. Both are negative for $r < r_c = d/(d - b)$, proving local asymptotic stability. For $r > r_c$ it is a saddle. At equality one eigenvalue is zero, so linearization alone is inconclusive. If $b \geq d$, both eigenvalues are negative for every $r > 0$.

For illustration, $a = 1, b = 1/2, d = 1, r = 3$ gives $r_c = 2$; the following physical-domain nullclines intersect at the DFE and one endemic equilibrium. The other algebraic intersection has $y > 1$ and is not shown.

$x' = 0$ (blue), $y' = 0$ (red)



□

- 14) Consider the 2nd-order ODE that describes the angle ϕ of a bead on a rotating hoop:

$$mr\ddot{\phi} + b\dot{\phi} + mg \sin \phi - mr\omega^2 \sin \phi \cos \phi = 0.$$

Here, m, b, g, r are positive parameters while $\omega \geq 0$.

- (a) By nondimensionalizing, show that the ODE can be written in the form

$$\varepsilon\phi'' + \phi' + \sin \phi - \gamma \sin \phi \cos \phi = 0.$$

for some dimensionless parameters ε and γ . Express ε and γ in terms of the parameters of the original system.

- (b) Under what assumptions about the original parameters can the system be well-approximated by the following 1D flow?

$$\phi' = -\sin \phi + \gamma \sin \phi \cos \phi$$

- (c) Assuming that $\gamma = 0$, find the fixed points of the above 1D flow and sketch a phase portrait.
 (d) Argue that a pitchfork bifurcation occurs at $\gamma = 1$. You may use graphical methods or any other valid approach to support your argument.

Solution. (a) Set $\tau = (mg/b)t$; primes below denote differentiation with respect to τ . Dividing the original equation by mg gives

$$\underbrace{\frac{m^2gr}{b^2}}_{\varepsilon} \phi'' + \phi' + \sin \phi - \underbrace{\frac{r\omega^2}{g}}_{\gamma} \sin \phi \cos \phi = 0.$$

Thus $\varepsilon = m^2gr/b^2$ and $\gamma = r\omega^2/g$.

(b) The overdamped approximation holds when $\varepsilon \ll 1$, equivalently $b^2 \gg m^2gr$, with γ of order one and after the fast velocity transient of dimensional duration about mr/b . Writing $v = \phi'$ gives $\varepsilon v' = -v - \sin \phi + \gamma \sin \phi \cos \phi$; its fast relaxation yields

$$\phi' = \sin \phi (\gamma \cos \phi - 1).$$

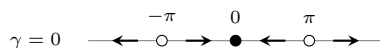
Initial angular velocity should be bounded on this scaling, or the initial relaxation layer must be accounted for. If γ is very large, the stronger separation condition $\varepsilon(1 + \gamma) \ll 1$ is a useful sufficient requirement; small ε alone need not control that regime.

(c) For $\gamma = 0$, $\phi' = -\sin \phi$. The equilibria are $k\pi$: even multiples are attracting ($f' = -1$), odd multiples repelling ($f' = 1$). On the circle there is one attracting point 0 and one repelling point π ; each interval $((2k - 1)\pi, (2k + 1)\pi)$ is the basin of $2k\pi$.

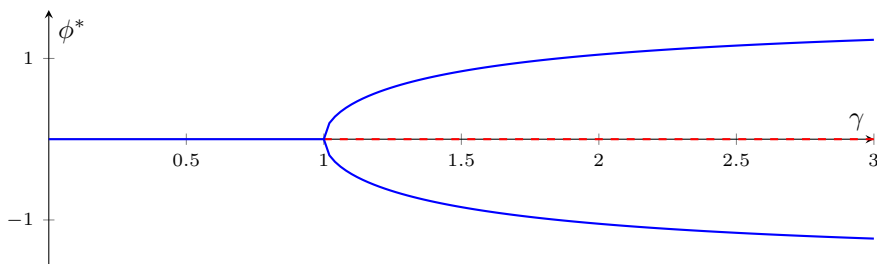
(d) For general $\gamma \geq 0$, the branches 0 and π modulo 2π always exist. At zero $f'(0) = \gamma - 1$; at π , $f'(\pi) = \gamma + 1 > 0$. For $\gamma > 1$ there are two additional branches

$$\phi_{\pm} = \pm \arccos(1/\gamma) \pmod{2\pi}, \quad f'(\phi_{\pm}) = \frac{1}{\gamma} - \gamma < 0.$$

Hence the stable origin loses stability and creates two stable branches: a **supercritical pitchfork** at $\gamma = 1$. With $\mu = \gamma - 1$, expansion gives $\phi' = \mu\phi - \frac{1}{2}\phi^3 + O(\mu\phi^3 + \phi^5)$ and $\phi_{\pm} \sim \pm\sqrt{2\mu}$. At $\gamma = 1$ the origin is still attracting, but convergence is algebraic rather than exponential.



Local bifurcation near $\phi = 0$



□

15) (25 points) Consider the 1D flow defined by

$$u' = r(u - 1) - u^2(u - 1).$$

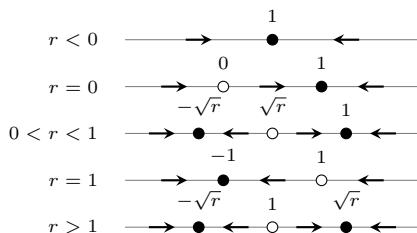
- Sketch all the qualitatively different vector fields that occur as the parameter r is varied.
- Find the value(s) of r at which bifurcations occur, and classify them as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
- Sketch a bifurcation diagram of fixed points u^* as a function of the parameter r . Be sure to indicate stability on your diagram.

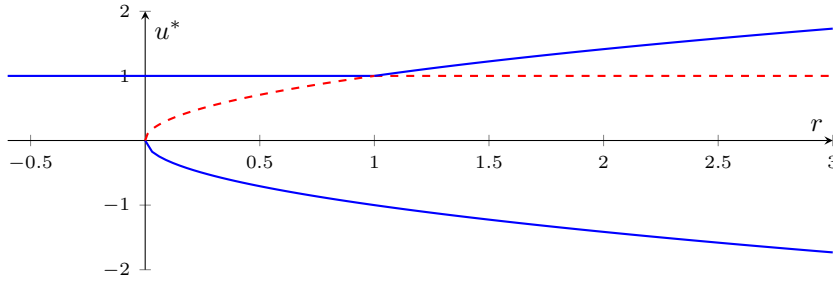
Solution. This is the same flow as problem 2, with the tasks split into three subparts. For completeness, the phase lines and bifurcation diagram are reproduced here.

Factor $f(u; r) = (u - 1)(r - u^2)$. The branches and their derivatives are

u^*	$f_u(u^*; r)$	stability
1	$r - 1$	attracting for $r < 1$, repelling for $r > 1$
$-\sqrt{r}$	$-2\sqrt{r}(\sqrt{r} + 1)$	attracting for $r > 0$
$\sqrt{r}$	$2\sqrt{r}(1 - \sqrt{r})$	repelling for $0 < r < 1$, attracting for $r > 1$

At $(u, r) = (0, 0)$, $f_r = -1$ and $f_{uu} = 2$, so a **saddle-node** creates the two square-root branches. At $(1, 1)$, putting $u = 1 + v$, $r = 1 + \mu$ gives $v' = v(\mu - 2v - v^2)$: a **transcritical** exchange of stability. At $r = 0$, $u = 0$ attracts from the left only; at $r = 1$, $u = 1$ attracts from the right only.





□

16) (25 points) Consider the system

$$\begin{aligned}\dot{x} &= -y \\ \dot{y} &= 2x(1 - x^2)\end{aligned}$$

- Find all fixed points of the system.
- Classify each of the fixed points you found in part (a), using eigenvalues of the linearized system. For fixed points with real eigenvalues, find the eigenvectors as well.
- Show that quantity $E = ax^2 + by^2 + x^4$ is conserved along trajectories for a suitable choice of a and b .
- Argue for the existence of periodic orbits in this model. You may use any valid method to support this claim.
- Sketch the nullclines and a plausible phase portrait for this system.

Solution. (a)–(b) The fixed points are $(0,0)$ and $(\pm 1,0)$, and

$$J = \begin{pmatrix} 0 & -1 \\ 2 - 6x^2 & 0 \end{pmatrix}.$$

At zero, the eigenvalues are $\pm i\sqrt{2}$, so the linearization is a center. The conserved quantity below proves it is also a nonlinear center. At either outer point, $\lambda = \pm 2$, with eigenvectors $(1, -2)^T$ for $\lambda = 2$ and $(1, 2)^T$ for $\lambda = -2$. Thus both outer fixed points are saddles.

(c) Differentiation yields

$$\frac{d}{dt}(ax^2 + by^2 + x^4) = xy[(-2a + 4b) + (-4b - 4)x^2].$$

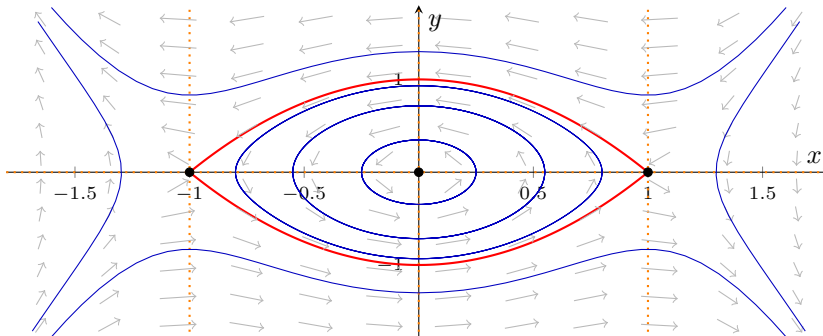
It vanishes identically when $b = -1$, $a = -2$. Hence $E = x^4 - 2x^2 - y^2$ is conserved. Equivalently use

$$H = -E/2 = \frac{1}{2}y^2 + x^2 - \frac{1}{2}x^4.$$

Indeed $\dot{x} = -H_y$ and $\dot{y} = H_x$.

(d) The origin is a strict local minimum of H . For $0 < h < 1/2$, the component of $H = h$ around zero is a smooth compact oval with no equilibria, so its flow is periodic. Its turning points satisfy $x^2 = 1 - \sqrt{1 - 2h}$. There are also unbounded components of these energy levels; the assertion concerns the inner oval only. At $h = 1/2$, the two inner curves $y = \pm(1 - x^2)$, $-1 < x < 1$, form a heteroclinic cycle between the saddles. This cycle bounds the region of periodic orbits.

(e) The nullclines are $y = 0$ and $x = 0, \pm 1$. Above the x -axis the flow moves left, below it right; $y' > 0$ for $x < -1$ or $0 < x < 1$, and $y' < 0$ for $-1 < x < 0$ or $x > 1$. The periodic orbits run counterclockwise. Red curves below are the two heteroclinic connections; orange dotted lines are nullclines.



□

- 17) (25 points) Consider the system 2D linear system $\dot{x} = Ax$, where $A = \begin{bmatrix} a & -1 \\ -1 & b \end{bmatrix}$ for parameters $a > 0$ and $b > 0$.

(a) Sketch a phase portrait and classify the fixed point at the origin for all possible qualitatively distinct cases as a function of parameters.

Note: Qualitatively accurate phase portraits are sufficient here, it is *not* necessary to compute eigenvectors of A .

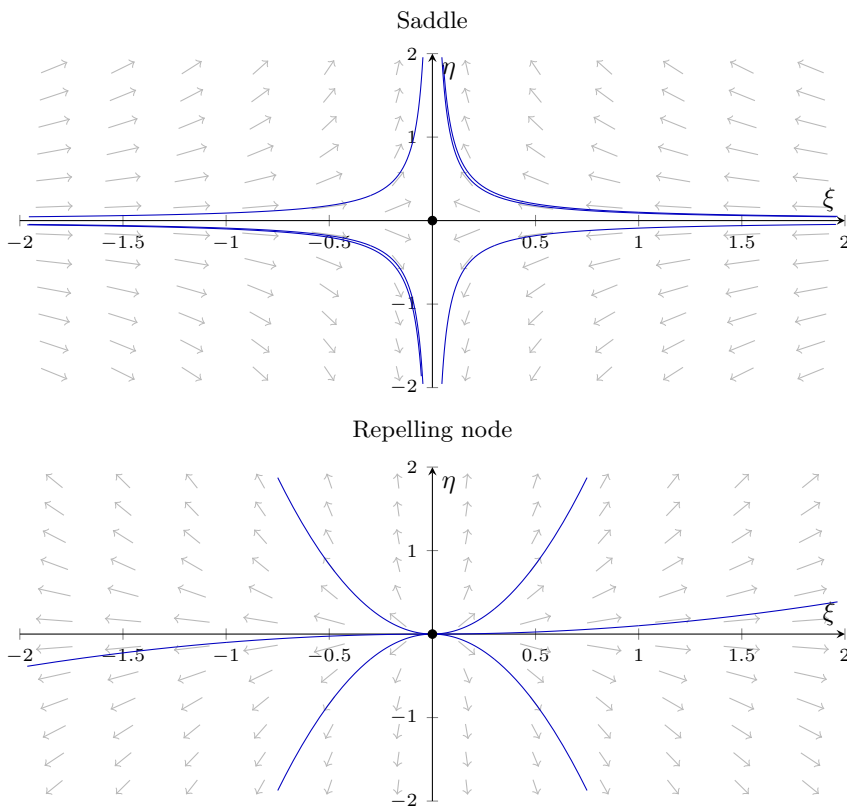
(b) Summarize these results by indicating the regions that you've identified in the first quadrant of the (a, b) -plane.

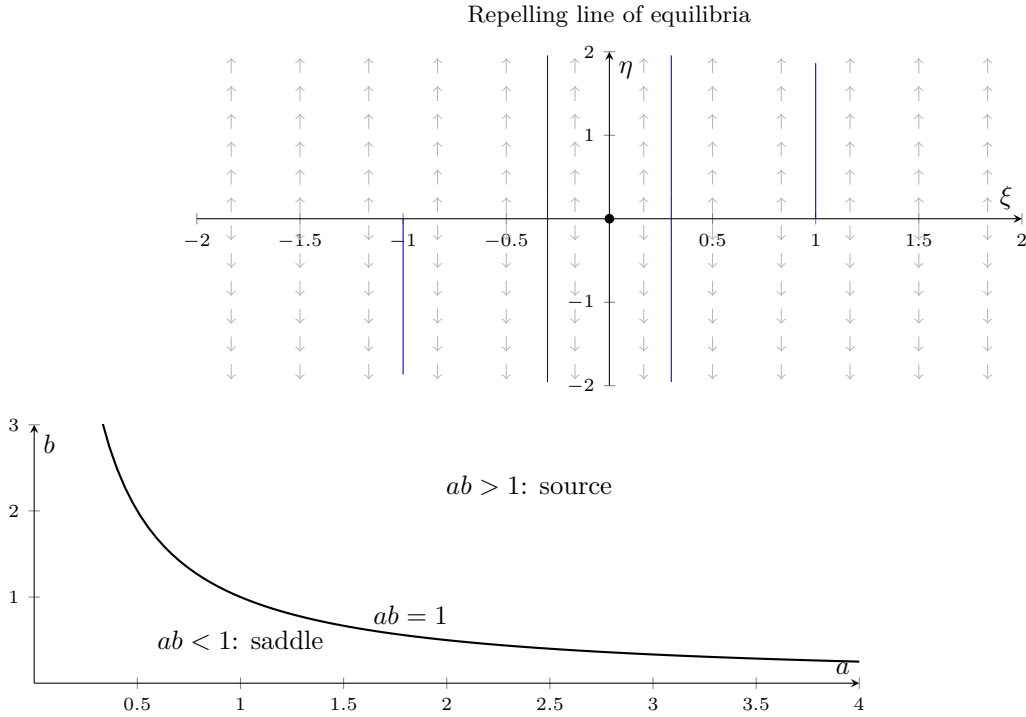
Solution. As in problem 8, interpret the printed equation as $\dot{\mathbf{x}} = A\mathbf{x}$. We have

$$T = a + b > 0, \quad D = ab - 1, \quad T^2 - 4D = (a - b)^2 + 4 > 0.$$

Thus the eigenvalues are real and distinct. For $ab < 1$ the origin is a saddle (one eigenvalue of each sign); for $ab > 1$ it is a repelling node (both positive). On $ab = 1$, the eigenvalues are 0 and $a + b$, so $y = ax$ is a line of equilibria that repels transversely. No attracting nodes or spirals occur in the first quadrant.

In eigenvector coordinates (ξ, η) , the portraits are respectively $\xi' = -\xi, \eta' = d\eta$; $\xi' = c\xi, \eta' = d\eta$; and $\xi' = 0, \eta' = d\eta$, with $c, d > 0$. The sketches below use these coordinates, so the coordinate axes are eigendirections. As requested, computing their original-coordinate eigenvectors is unnecessary.





□

- 18) (25 points) Consider the 2nd-order ODE that describes the angle ϕ of a bead on a rotating hoop:

$$mr\ddot{\phi} + b\dot{\phi} + mg \sin \phi - mr\omega^2 \sin \phi \cos \phi = 0.$$

Here, m, b, g, r are positive parameters while $\omega \geq 0$.

- (a) By nondimensionalizing, show that the ODE can be written in the form

$$\varepsilon \phi'' + \phi' + \sin \phi - \gamma \sin \phi \cos \phi = 0.$$

for some dimensionless parameters ε and γ . Express ε and γ in terms of the parameters of the original system.

- (b) Under what assumptions about the original parameters can the system be well-approximated by the following 1D flow?

$$\phi' = -\sin \phi + \gamma \sin \phi \cos \phi$$

- (c) Assuming that $\gamma = 0$, find the fixed points of the above 1D flow and sketch a phase portrait.
 (d) Argue that a pitchfork bifurcation occurs at $\gamma = 1$. You may use graphical methods or any other valid approach to support your argument.

Solution. This is the same bead-on-hoop model as problem 14. The full nondimensionalization and bifurcation analysis follow.

(a) Set $\tau = (mg/b)t$; primes below denote differentiation with respect to τ . Dividing the original equation by mg gives

$$\underbrace{\frac{m^2 gr}{b^2}}_{\varepsilon} \phi'' + \phi' + \sin \phi - \underbrace{\frac{r\omega^2}{g}}_{\gamma} \sin \phi \cos \phi = 0.$$

Thus $\varepsilon = m^2 gr/b^2$ and $\gamma = r\omega^2/g$.

(b) The overdamped approximation holds when $\varepsilon \ll 1$, equivalently $b^2 \gg m^2 gr$, with γ of order one and after the fast velocity transient of dimensional duration about mr/b . Writing $v = \phi'$ gives $\varepsilon v' = -v - \sin \phi + \gamma \sin \phi \cos \phi$; its fast relaxation yields

$$\phi' = \sin \phi (\gamma \cos \phi - 1).$$

Initial angular velocity should be bounded on this scaling, or the initial relaxation layer must be accounted for. If γ is very large, the stronger separation condition $\varepsilon(1 + \gamma) \ll 1$ is a useful sufficient requirement; small ε alone need not control that regime.

(c) For $\gamma = 0$, $\phi' = -\sin \phi$. The equilibria are $k\pi$: even multiples are attracting ($f' = -1$), odd multiples repelling ($f' = 1$). On the circle there is one attracting point 0 and one repelling point π ; each interval $((2k - 1)\pi, (2k + 1)\pi)$ is the basin of $2k\pi$.

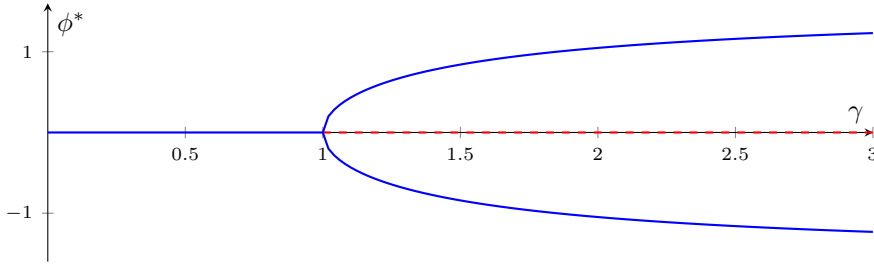
(d) For general $\gamma \geq 0$, the branches 0 and π modulo 2π always exist. At zero $f'(0) = \gamma - 1$; at π , $f'(\pi) = \gamma + 1 > 0$. For $\gamma > 1$ there are two additional branches

$$\phi_{\pm} = \pm \arccos(1/\gamma) \pmod{2\pi}, \quad f'(\phi_{\pm}) = \frac{1}{\gamma} - \gamma < 0.$$

Hence the stable origin loses stability and creates two stable branches: a **supercritical pitchfork** at $\gamma = 1$. With $\mu = \gamma - 1$, expansion gives $\phi' = \mu\phi - \frac{1}{2}\phi^3 + O(\mu\phi^3 + \phi^5)$ and $\phi_{\pm} \sim \pm\sqrt{2\mu}$. At $\gamma = 1$ the origin is still attracting, but convergence is algebraic rather than exponential.

$$\gamma = 0 \quad \leftarrow \text{---} \overset{-\pi}{\circ} \text{---} \overset{0}{\bullet} \text{---} \overset{\pi}{\circ} \text{---} \rightarrow$$

Local bifurcation near $\phi = 0$



□